SUMMARY AT THE BOTTOM

Strain is a key concept in engineering and physics that describes how a material deforms in response to applied stress. The ratio of an object's change in length or shape to its original length or shape is known as strain. In many disciplines, including structural engineering, mechanical engineering, and materials science, an understanding of strain is essential because it determines how a material will behave under load, and enables the design of structures that can withstand the expected service stresses.

A strain gauge is a sensor that measures the amount of strain (deformation) in an object by converting mechanical force into a measurable change in electrical resistance. When external forces such as *pressure*, *tension*, *compression*, or *weight* are applied to a stationary object, they create internal resisting forces known as **stress** and physical deformation known as **strain**.

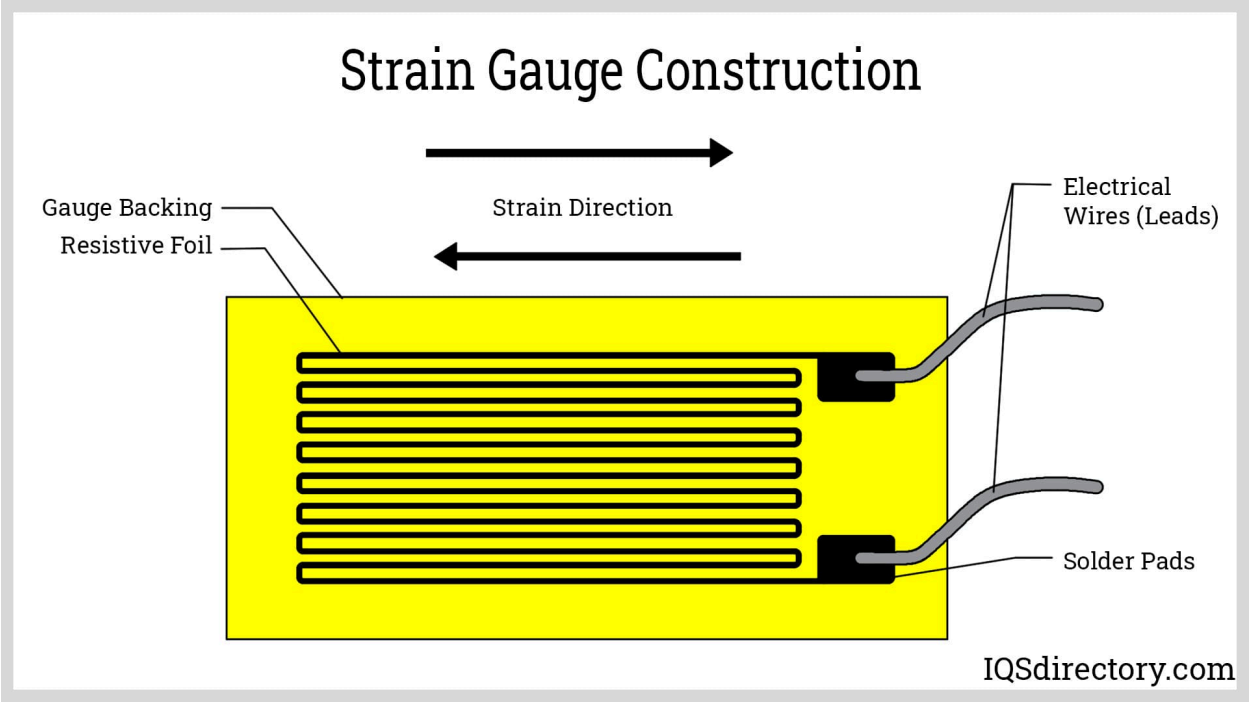
The working principle of a strain gauge is based on the piezoresistive effect, or the tendency of a material's electrical resistance to change when it is stretched or compressed.

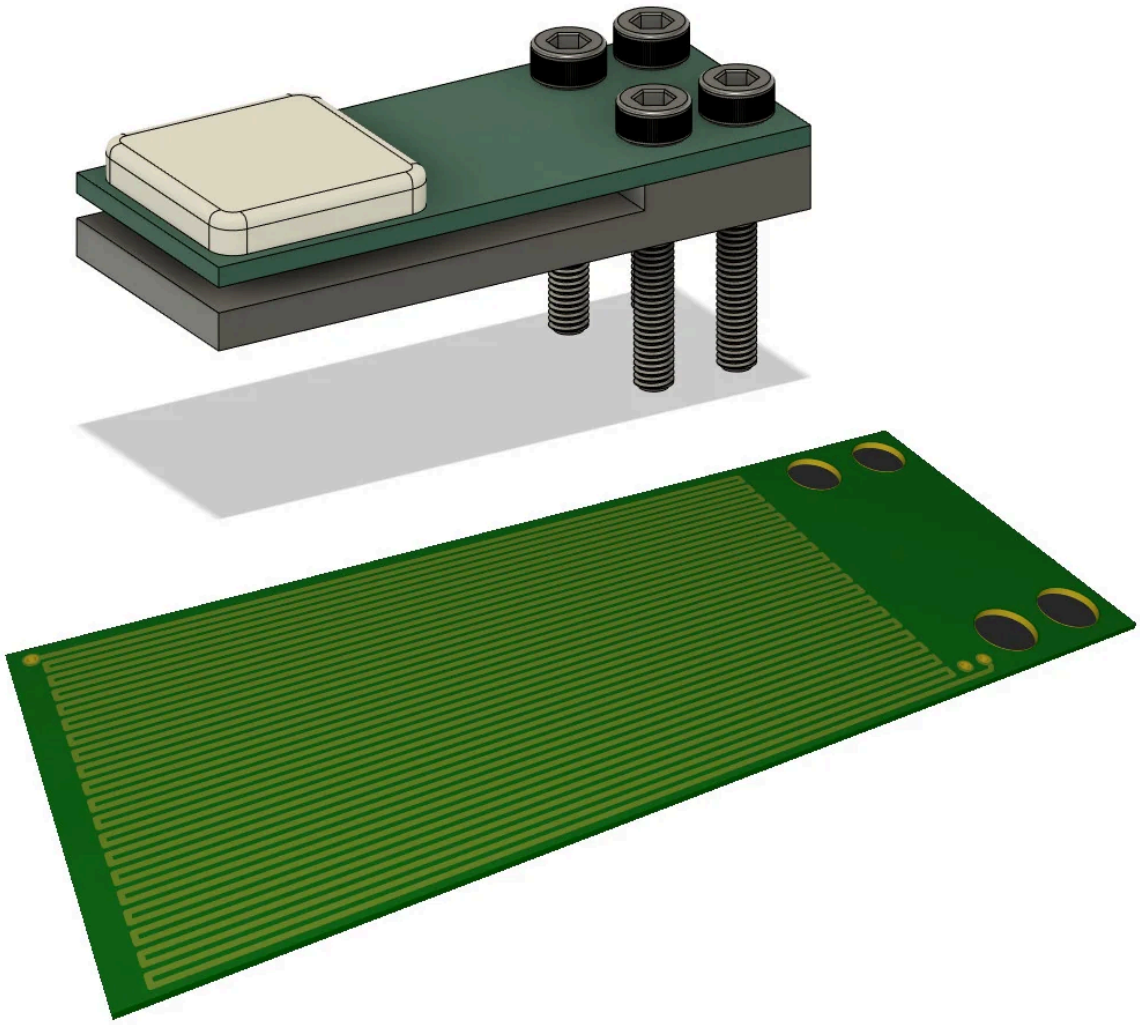
A strain gauge is an instrument used to measure the strain or deformation of a component. A number of tiny metal foil gauges are bonded to an object and an electrical current is passed through them. As the object, and hence the gauges, deform, their electrical resistance changes. When this minute signal (typically in the region of millivolts) is amplified, it can be measured and logged with a data logger. A strain gauge and amplifier arrangement can also be referred to as a load cell. When the mechanical properties of the gauged material are known, the measured deformation can be used to calculate applied loads and torques. Within motorsport this technology has been applied to measure many characteristics of a racecar, including:

- suspension loads
- driveshaft and propshaft torque
- gear lever force
- pedal forces
- steering system torques and loads

usually pcb is connection of wires (trace), pad, third thing connecting inputs and output ports but for strain gauge , pcb *is* the component

basically a long wire squiggling
look at strain gauge design
 i need to do the math to figure out the logistics of the trace
and theres input and output terminal that are for measuring resistance, but thats outside scope


Strain Gauge Construction
Strain Direction
Gauge Backing
Resistive Foil
Electrical
Wires (Leads)
Solder Pads
IQSdirectory.com



R = (pL)/A = (1.68e-8 * L)/A

A = cross sectional area
R = resistivity (ohms)
p = copper resistivity = 1.68e-8 ohm m
L = length

also not exactly 350 ohms
temperature can affect?
Trace Width Tolerance - ±20%

Minimum Trace Width/Spacing (1 oz)
aka: width of traces
traces are basically wires but easier to manufacture rather than soldering on copper wires in the same config according to reddit
① 12 µm (0.33 oz) copper: 3/3 mil (absolute limit 2/2 mil – avoid if possible)
3/3 mil
mil = 0.001 inch
the first number of 3/3 means trace width
the second number means trace spacing (how far apart the traces have to be)
3/3 = 0.0762 millimeters / 0.0762 millimeters

② 18 µm (0.5 oz) copper: 3.5/3.5 mil

③ 35 µm (1 oz) copper: 4/4 mil

lets pick the smallest one for more traces on board? ? ?

TRACE WIDTH: 0.0762 millimeters (or 0.003 inches)
TRACE SPACING: 0.0762 millimeters (or 0.003 inches)

alternatively? they say 2/2 is absolute limit so maybe do 2.5/2.5?
TRACE WIDTH: 0.0635 millimeters(or 0.0025 inches)
TRACE SPACING: 0.0635 millimeters (or 0.0025 inches)
SO
$R = (pL)/A = (1.68e\text{-}8 * L)/A$

A = cross sectional area
R = resistivity (ohms)
p = copper resistivity = 1.68e-8 ohm m
L = length

cross sectional area of a trace is rectangular apparently

> Unlike wires with circular cross-sections, PCB traces have rectangular cross-sections, making their current capacity calculations more complex.

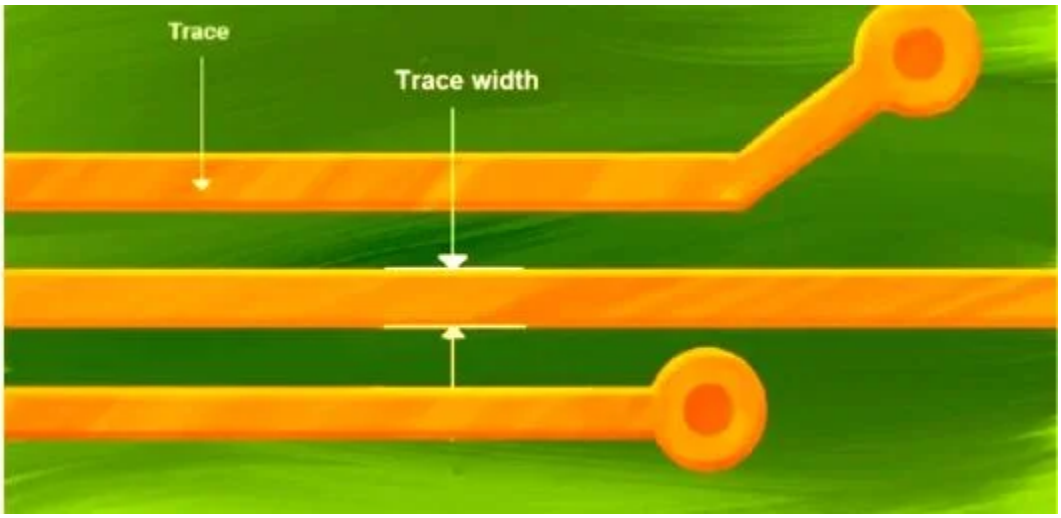
https://www.syspcb.com/pcb-blog/knowledge/pcb-trace-width-copper-thickness-and-current-relationship-a-design-guide.html

PCB manufacturers specify copper thickness using the term **"copper weight"** measured in ounces (oz). This historical measurement refers to the weight of copper spread evenly over one square foot of area.

For a **20°C temperature rise**, these values can be multiplied by approximately 1.5, and for **30°C rise**, by approximately 1.8.

so temperature increases Current Capacity?? ?

6.**Account for manufacturing tolerances** – Add 10-20% safety margin to calculated widths to accommodate etching variations



rectangular cross sectional area: trace width * copper thickness (0.012 millimeters)
so for 2.5/2.5 mil,
A = 7.62e-10 m
length:
L = AR/p
L = 350 * 7.62e-10 / (1.68e-8) = 15.87500 m

length of longwise: choosing 0.1 m
so we need

x runs and x-1 gaps between them
so x + x-1 = 2x-1 widths total since 2.5 == 2.5 of 2.5/2.5 => 0.0635 mm

Width = 0.0635 * (2x-1)
bends are circular , r = 1mm ????? length of bend = 3.1415mm and x-1 bends
0.001pi * (x-1) + 0.1(x) = 15.87500
x(0.1 + 0.001pi) = 15.875 + 0.001pi > x =15.8781416/ 0.103141593 = 153.945088
so x = 154
width = 0.0635*(2x-1) = 307 * 0.0635 =19.495 mm =>

so the flexible pcb is 0.01949 m wide and 0.1 m long ??
0.767322835 inches wide and 3.93700787 inches long

CHANGE BENDS TO R = 0.0635 mm according to luca
length of bend = 0.0635 mm * pi = 0.000199491134 m ~~ = 0.0002 m
0.0002 * (x-1) + 0.1x = 15.875
0.1002x = 15.8752 => x = 158.43513 ~~ = 158
so 158 runs
width = 0.0635 mm * (158 * 2 - 1) = 20.0025 millimeters

SO

EXPLANATION

Which copper wire/trace kind to pick ? Flexible pcb website JRL says : there are 3 options
I picked the smallest one (idrk why, maybe because smallest can fit more on one pcb? the larger ones are for higher power /current applications??? idk) (but thats the one i picked)
this is the details of the smallest one: "① 12 µm (0.33 oz) copper: 3/3 mil (absolute limit 2/2 mil – avoid if possible)"
3/3 mil -> 3mil trace width, 3mil trace gaps (between parallel/adjacent traces)
okay but they said 3/3 mil is the smallest and the absolute smallest is 2/2 mil which should be avoided probably because it causes issue
So why not do 2.5/2.5 mil ? as a middle ground ?
that results in
TRACE WIDTH = 2.5 mil = 0.0635 mm
TRACE GAPS WIDTH = 0.0635 mm
TOTAL LENGTH OF TRACE TO MAINTAIN ~350OHMS RESISTANCE = 15.875m
(I calculated this using:

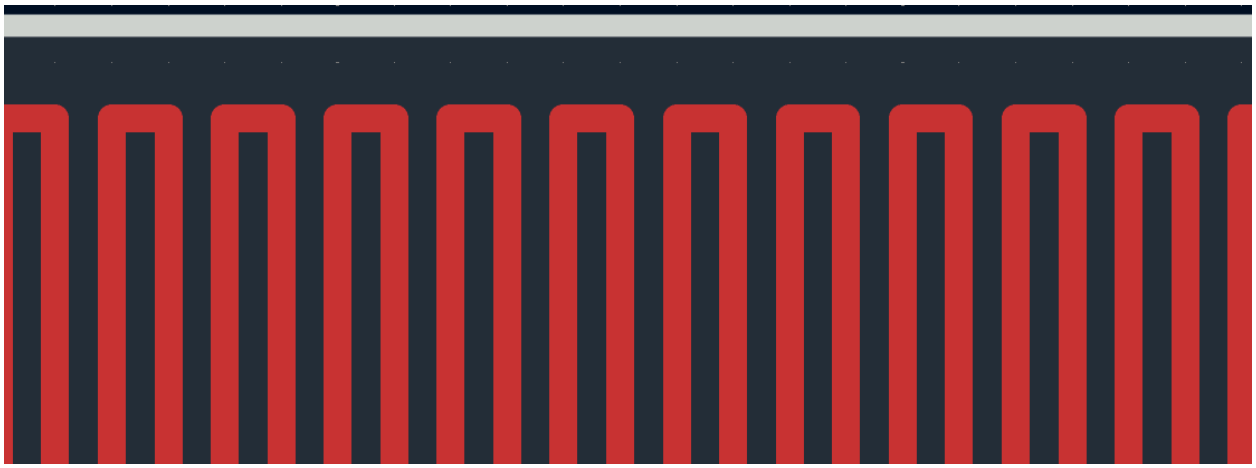
- TRADE WIDTH and TRACE GAPS WIDTH
- COPPER THICKNESS (this kind of trace is 12um aka 0.012 mm)
- to obtain the cross sectional area, A , for a trace its rectangular , so 0.012mm * 0.0635 mm -> A = 7.62e-10 m
- Resistance = 350Ohms (given)
- material resistivity of copper = 1.68e-8 ohm meters (obtained online)

RESULTING IN L = 15.875 m

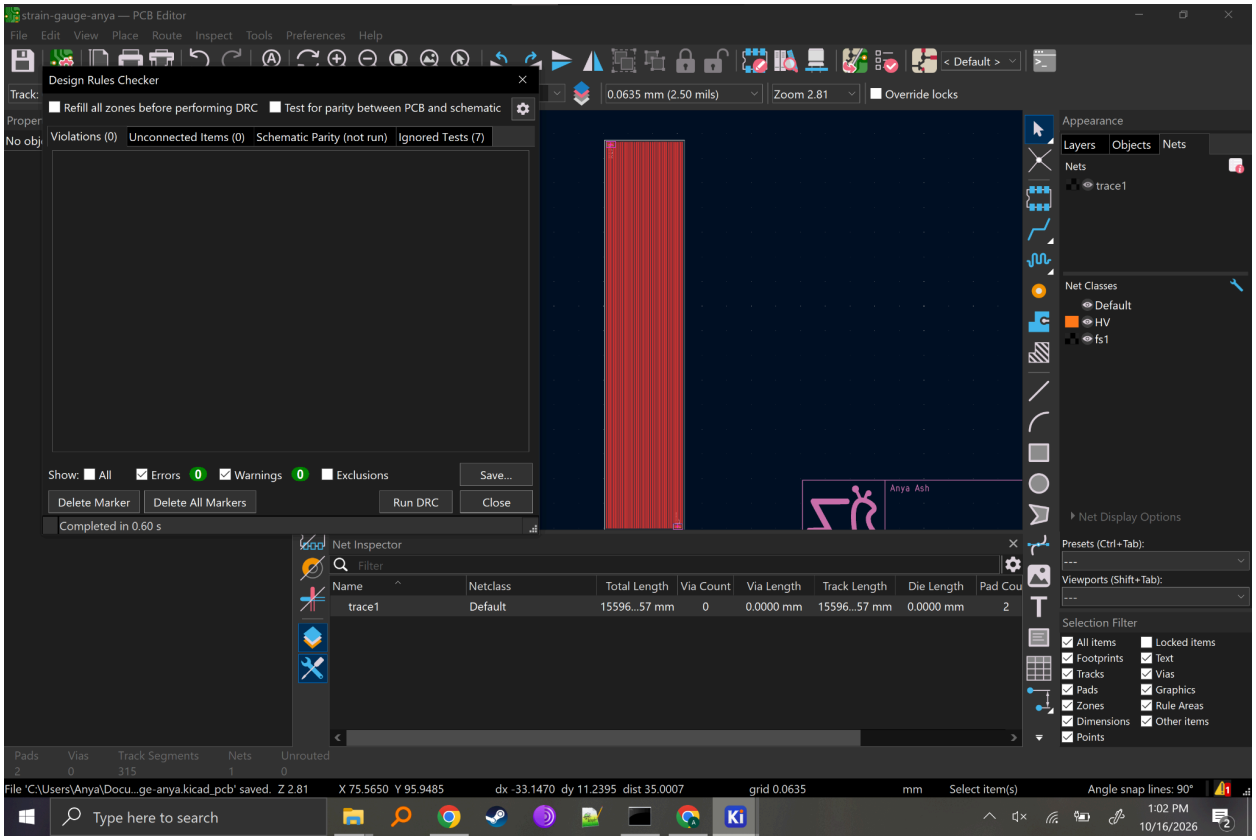
NOW FOR THE Flexible pcb
LENGTH = 0.1 m = 100 mm (I randomly/arbitrarily chose this)
WIDTH = 0.0200025 m = 20.0025 mm (I calculated this based on the total length of trace, accounting for how many runs (aka individual lines/parts of the squiggle) and how many bends between/connecting them)
TOTAL NUMBER OF RUNS = 158.
SOLDERING PADS: 2 on each end of the trace, and they are 2 mm x 2 mm.
also the bends are round, half circles. not diagonal.

according to luca: flexible PCB has solder mask layer, copper trace layer, and fiber glass layer. if you just put the copper traces, there is no hole to solder to. so you need to add pads (recommended 2 mm x 2mm) to solder to.
I did not account for the pads in the length calculation but since they are so small (8sq mm of a 2000 sq mm pcb aka 0.4%) - are pads a component next to the trace at the ends or is it on top of the trace? if just on top, then it doesn't matter how big they are . if not on top, they are relatively small that removing that much trace wont make a big difference on the 350 ohms number.

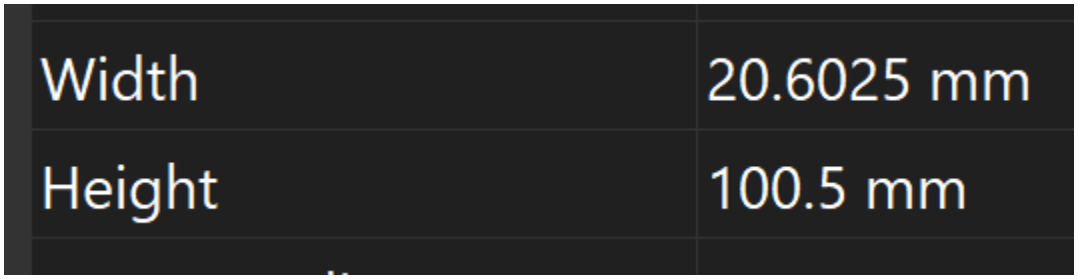
kicad
actual track segments i could fit on the board: 157 (idk where the discrepancy came from)
which i guess worked out fine because i think i need an odd number of runs ?



also ended up just doing this instead of semi circle bends

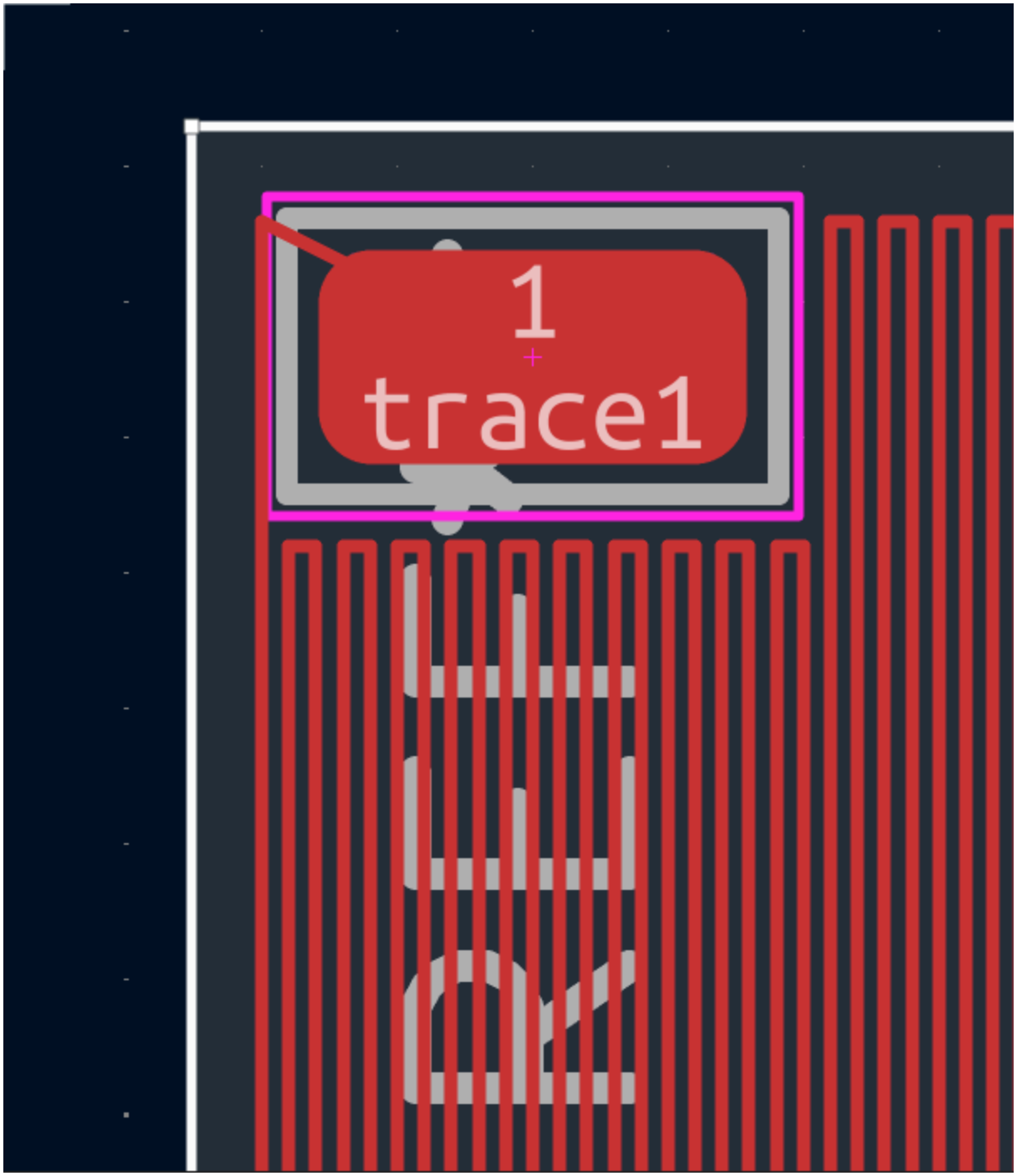


finished design and it has 0 errors 0 warnings



i ended up extending the length of the board by 0.5 mm and width of the board by 0.6 mm to meet the copper to board edge >= 0.3 mm requirement (the trace didnt touch to the edge of the height before so i didn’t need to extend by 0.6mm just 0.5mm)

total trace length = ~15596 mm
which with A = 7.62e-10 m, p = 1.68e-8 ohm m, R = pl/A => Ideal resistivity = 343.848819 ohms
344 ohms is close enough to 350.



the soldering pads i used are

SolderWirePad_1x01_SMD_1x2mm	Wire Pad, Square, SMD Pad, 1mm x 2mm

because they were the smallest wire pad available
i also realized they are pretty big so that is why i wasnt able to get the full length (15.875)
but yeah i would consider the design a success because it is really close to 350 ohms.
link to kicad files download